

Tuesday, 17/2/2026

## Carbon Dioxide and Global Warming A Story of Coupled Molecular Dynamics

The Earth is warming up, and the cause is said to be the increase in greenhouse gases like carbon dioxide ( $\text{CO}_2$ ) in the atmosphere. Infrared light is the primary type of radiation responsible for heating, and more infrared radiation in our atmosphere will mean higher temperatures. When sunlight reaches Earth, the surface absorbs some of the light's energy and re-radiates it as infrared waves, which we feel as heat. (Hold your hand over a dark rock on a warm sunny day and you can feel this phenomenon for yourself.) These infrared waves travel up into the atmosphere and will escape back into space if unimpeded.

Oxygen and nitrogen don't interfere with infrared waves in the atmosphere. That's because molecules are picky about the range of wavelengths that they interact with. For example, oxygen and nitrogen absorb energy of radiation having frequency around  $10^{20}$  Hz (which corresponds to their resonant frequencies, akin to normal modes), whereas infrared radiation is around  $10^{14}$  Hz or so (there is a wide range of frequencies for infrared, such as in between  $3 \times 10^{11}$  Hz -  $4 \times 10^{14}$  Hz, but let's focus our attention around  $10^{14}$  Hz). Those ranges don't overlap, so to oxygen and nitrogen, it's as if the infrared waves don't even exist; they let the waves (and heat) pass freely through the atmosphere.

There seems to be an obvious correlation between rise in average global temperatures in the last century, and the amount of carbon dioxide present in the atmosphere. Have a look at <https://climate.nasa.gov/vital-signs/global-temperature/> to see the trend in global temperature rise, and then look at <https://climate.nasa.gov/vital-signs/carbon-dioxide/> to see how  $\text{CO}_2$  levels have risen in the atmosphere over the years.

The question, then is, what is different about  $\text{CO}_2$ ? Is  $\text{CO}_2$  able to "soak up" this infrared energy which is then emitted and returned back to the Earth as heat, contributing to the "greenhouse effect?" **Your mission, should you choose to accept it, is to analyze the properties of carbon dioxide using techniques of coupled oscillators you learned in this class (and other data provided in the question), and explain whether  $\text{CO}_2$  can be contributing to the greenhouse effect?**

To help you get started, here is some helpful information about the carbon dioxide molecule, and the chemical bonds that constitute it. Carbon dioxide is a linear, triatomic molecule with a central carbon atom, connected to an oxygen atom on each side. It can classically be modeled as a system with a central mass  $m = 12$  AMU (for the carbon) connected by equal springs of spring constant  $k$  to two masses  $M = 4m/3 = 16$  AMU (for the oxygen) constrained to move only along the line joining their centers. The harmonic vibrations of carbon dioxide give it its absorption properties.

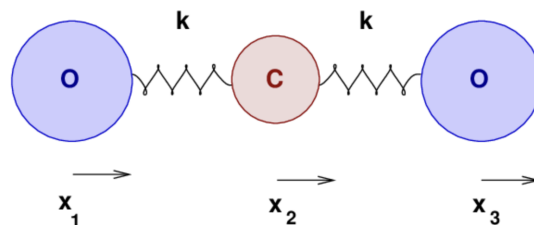
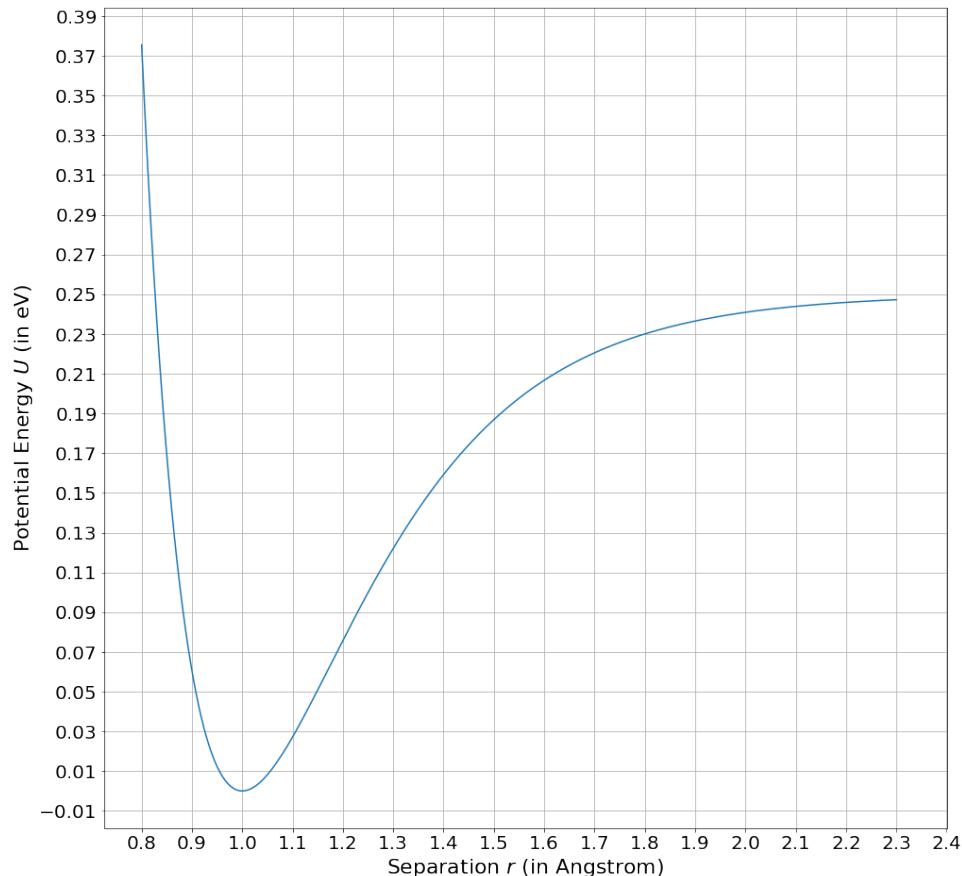


Figure 1: Schematic atomic bonds making up the  $\text{CO}_2$  molecule. Credit: RIT

- a Write down the equations of motion ( $F = ma$ ) for each of the three atoms involved in  $\text{CO}_2$ . You will find it to be a coupled set of equations. Write it as a  $3 \times 3$  matrix equation similar to the way we have been doing in class.

- b How many normal modes do you expect this system to have?
- c Set up and solve for the normal modes of oscillations for  $\text{CO}_2$ . Find the frequencies of oscillations of these modes in terms of  $m$ ,  $M$  and  $k$ .
- d You would notice that one of these normal modes does *not* oscillate. Can you physically explain this? Use the eigenvector to understand it better.

Now let us get a bit more quantitative. The carbon - oxygen,  $\text{C} - \text{O}$ , bond energy (which is the potential energy due to the attractive and repulsive forces between the atoms) has been studied and measured, and it looks like the following graph. Remember  $1\text{eV} = 1.6 \times 10^{-19}\text{ J}$ , and  $1\text{\AA}$  (Angstrom) is  $10^{-10}\text{ m}$ . The separation  $r$  is the distance between the carbon and the oxygen atoms.



Such potentials are often modeled by the interatomic “Morse” potential, which has the following form (you also met this potential during Tutorial 2),

$$U(r) = D \left( 1 - e^{-a(r-b)} \right)^2, \quad (1)$$

where  $D, a, b$  are positive constants.

- e Use the above potential energy plot and your understanding of small oscillations around local stable equilibrium to estimate the bond strength  $k$  of the  $\text{C} - \text{O}$  bond. Note the plot is drawn *to scale*. There are multiple ways to do this, and I encourage you to discuss with Prof. Singh and TAs as you approach this.
- f Now, give a numerical estimate for the normal mode frequencies you found earlier in this question. In case you have forgotten  $1\text{ AMU} = 1.6 \times 10^{-27}\text{ kg}$ .
- g Based on the fact that normal mode frequencies are the system’s natural frequencies of oscillations (it if, after all, the collective vibration of the molecule), determine whether  $\text{CO}_2$  can absorb infrared radiation and contribute to global warming.